

Integration of onboard sensors and Local Area DGNSS to Support High Integrity Unmanned Aerial Vehicles (UAV) Navigation

Jinsil Lee, Minchan Kim, and Jiyun Lee
Korea Advanced Institute of Science and Technology

ABSTRACT

A Local-Area Differential GNSS (LAD-GNSS) provides high integrity and accuracy for Unmanned Aerial Vehicles (UAVs). The integration of onboard sensors to the LAD-GNSS can support continuous and reliable navigation by overcoming GNSS signal vulnerability. This paper proposes the concept of a high integrity navigation architecture into which the LAD-GNSS and onboard sensors are integrated using an extended Kalman filter (EKF). Key components of the integrated system developed in this work, include its integrity/continuity requirement design, fault-monitoring strategies, and the corresponding protection level (PL) computation formulations.

In this paper, the total integrity requirements are divided into mainly the following four hypotheses: a fault-free condition, a reference receiver failure, all other failures of LAD-GNSS, and an onboard sensor failure. According to the integrity allocation, fault monitoring strategies and PL computation formulations were proposed. The ground monitor of the LAD-GNSS and the onboard Receiver Autonomous Integrity Monitoring (RAIM) are utilized to monitor a single satellite failure sequentially, and an innovation-based fault-monitoring method is applied to monitor onboard sensor failures. To evaluate the performance of the integrated system, the VPLs under the four hypotheses were calculated using data collected from flight experiments which used an Inertial Navigation Sensor (INS) as an onboard sensor. The results show that the PL for the INS failure hypothesis has the largest vertical protection level (VPL) for the most of the operation time and it suitably bounds vertical position errors during a flight test.

1. INTRODUCTION

As the demand for autonomous Unmanned Aerial Vehicle (UAV) applications increase, reliable navigation sensors and ground infrastructures will become more vital to enable UAVs to conduct various missions with minimal human intervention. For safe UAV operations, all UAVs must guarantee that their navigation solution is adequately being protected by an integrity assurance system. In

previous work, Local-Area Differential GNSS (LAD-GNSS), which is a simplified architecture of the Ground-based Augmentation System (GBAS) particularly for UAVs that emphasizes low costs for commercial applications and high integrity, has been proposed [1, 2]. The ground facility broadcasts differential GNSS corrections and integrity information to meet the required levels of integrity (comparable to those of GBAS Category I – III operations) by monitoring navigation faults at the reference station. Using the navigation system error models whose parameters are included in the integrity information, UAVs can compute their conservative position error bounds (i.e., protection levels (PL)). While the LAD-GNSS can be the primary source of navigation for UAVs operating in controlled areas where no excessive GNSS signal interference or blockages exist, the system requires the aid of other navigation sensors to provide continuous and reliable navigation considering a large variety of UAV operating environments. In addition, a higher rate of solutions from integrated sensors than the 2 Hz solution of LAD-GNSS is appropriate to support dynamic operations of UAVs. As an integrated system should also guarantee a very high probability that UAVs do not collide with the ground or potentially with other vehicles, such a system requires levels of integrity and continuity similar to what is achieved by LAD-GNSS for safe UAV operations. For this reason, the integrity architecture designed for the existing monitoring system should be extended to take into account the contribution of the additional sensors.

This study therefore proposes a high integrity navigation architecture for integrated LAD-GNSS/onboard navigation sensor systems by redesigning the existing LAD-GNSS integrity architecture. The proposed architecture assumes the onboard sensor failure as an additional failure hypothesis, as other types of failure hypothesis are considered in LAD-GNSS. With the integration of the onboard sensors to the LAD-GNSS system, several key components in the system should be redesigned. These include an integrity/continuity requirement fault-trees design, fault-monitoring strategies, and corresponding PL computation formulations for each failure hypothesis. In this study, Inertial Navigation Sensor (INS) is considered as an onboard sensor, and an

extended Kalman filter (EKF) is adopted as a navigation filter for integrating multiple sensors. To verify the proposed concept, PLs of an EKF-generated user position are derived using collected data from actual flight tests and are then compared with true trajectories.

2. DESIGN OF THE ARCHITECTURE FOR THE LAD-GNSS AND ONBOARD SENSORS INTEGRATED SYSTEM

This section introduces newly designed components that should be considered when aiding sensors are integrated into the existing LAD-GNSS.

2.1 Revised System Requirement Fault-Trees: Integrity/Continuity Requirements

This study designs a revised integrity/continuity allocation for the LAD-GNSS/onboard sensor integrated system to support UAVs that navigate with different types of sensors. Figure 1 shows the integrity fault tree of the revised system. The total system signal in space (SIS: encompassing GPS satellites and GBAS VDB) integrity risk requirement of 2×10^{-7} per approach is allocated to different types of failure modes. The total system integrity requirement remains unchanged from that of the LAD-GNSS to ensure performance capabilities identical to those of the LAD-GNSS. Instead, sub-allocations are adjusted by the inclusion of an onboard sensor failure hypothesis. In the re-designed integrity fault tree, one half of the total is allocated to the onboard sensor failure hypothesis. In response, the remaining 50% of the total integrity risk not allocated to the onboard sensor failure hypothesis (or 1×10^{-7}) is re-allocated to existing fault modes of the LAD-GNSS. The allocations to existing types of failure hypotheses of the LAD-GNSS/onboard sensor integrated system become tighter than those of the LAD-GNSS. Both the nominal hypothesis and the single reference receiver fault (H1) hypothesis are correspondingly reduced to 1×10^{-8} . The remaining integrity risk from the existing failure type of hypothesis ($10^{-7} - 2 \times 10^{-8} = 8 \times 10^{-8}$ per approach) is allocated to the remaining single-satellite vehicle (SSV) failures and all

other failure conditions of LAD-GNSS. As the risk allocation become tight, the onboard Receiver Autonomous Integrity Monitoring (RAIM) monitor is designed to compensate for the tight allocation. The onboard RAIM monitor applies the a-prior probability of failure as much as is assured by LAD-GNSS monitor, resulting in a reduced missed detection probability of failure. This monitoring concept can reduce the corresponding PLs. This will be explained in detail in Section 2.3.

The Continuity requirement is also reallocated, as shown in Figure 2. The total continuity requirement is also identical to that associated with the LAD-GNSS. In this revised fault tree, approximately 86% of the total continuity risk is allocated to the possibility of PLs exceeding the required alert limit, and they are subdivided into three categories: those with configuration changes, those without configuration changes, and those due to integrity warnings from the onboard sensors. The configuration changes are defined as unexpected events that cause the loss of one or more ranging sources or reference receivers. Therefore, the continuity risk to the integrity warning from the onboard RAIM, for detecting the SSV failures, is assigned in the hypothesis pertaining to PLs with configuration changes, while that of integrity warnings from onboard sensors is allocated separately. To allocate some portion of the continuity risk to the onboard RAIM and onboard sensor monitor, the continuity risk which was unallocated in the LAD-GNSS (2×10^{-7}) and the partial continuity risk allocated for ground monitoring (8×10^{-7}) are reallocated to the onboard RAIM and the onboard sensor monitor. As a result, continuity risk of 5×10^{-7} is allocated to onboard RAIM monitoring and onboard sensor monitoring, respectively. The continuity risks associated with hardware restrictions remain unchanged.

In this study, we focus on failure scenarios that must be revised when the onboard sensor failure hypotheses are included. The hypotheses consist of the fault-free ('H0'), the reference receiver failure ('H1'), all other failures of LAD-GNSS ('H2'), and the onboard sensor failures hypothesis (denoted as 'H3' in this paper).

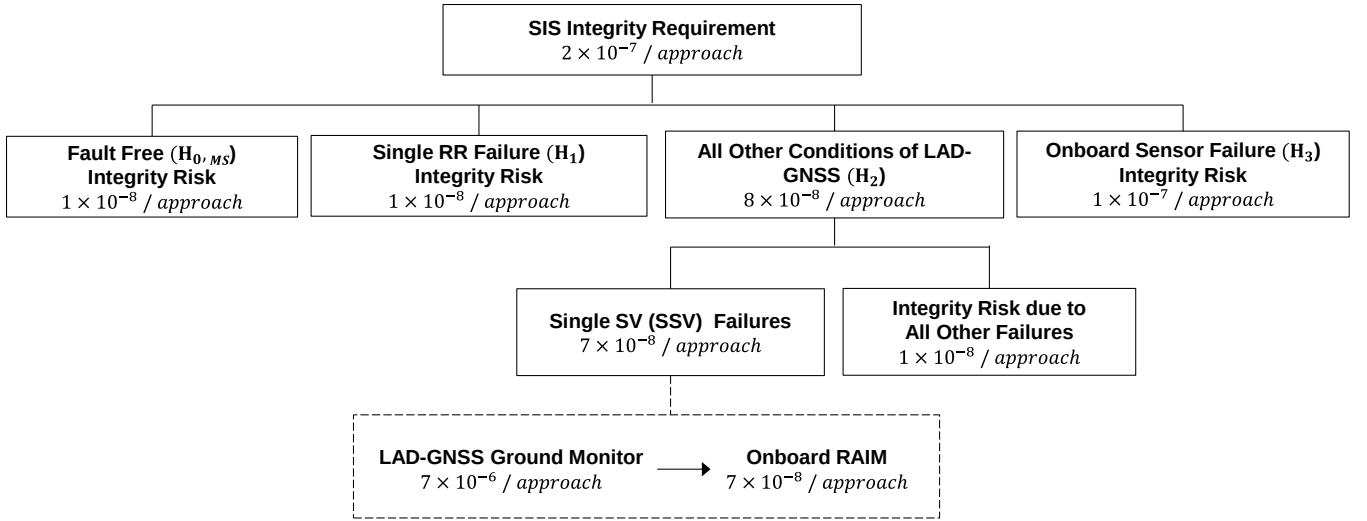


Figure 1. Integrity risk allocation for the LAD-GNSS/onboard sensor integrated system

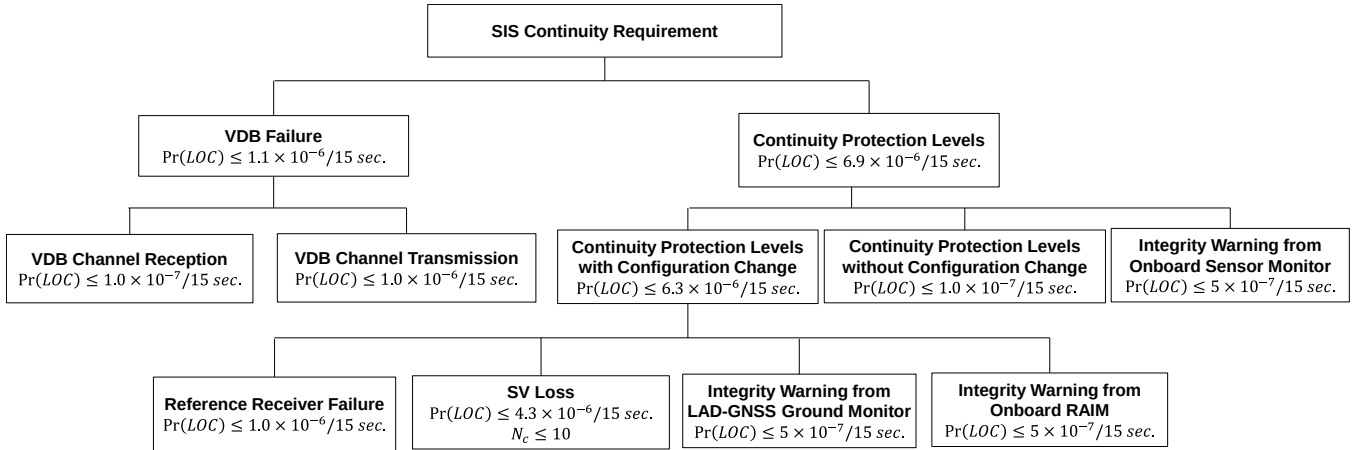


Figure 2. Continuity risk allocation for the LAD-GNSS/onboard sensor integrated system

2.2 Fault-Free Hypothesis: H0, MS

In the fault-free hypothesis of the LAD-GNSS/onboard sensor integrated system, the nominal vertical protection level (VPL) is evaluated with all of the sensor inputs of the LAD-GNSS and onboard sensors. The original expression of the fault-free hypothesis, H0, is then revised to 'H0, MS' where MS refers to multi-sensor. For the integration of multi-sensor inputs, the EKF navigation filter estimates the errors or deviations of the a priori and a posteriori user positions, velocities, and attitudes relative to their true values. In this study, specific forces and angular rates from an INS are applied to compute the predicted state vector estimates ($\hat{x}_k^-$). The predicted state vector estimates are updated by an input measurement from the LAD-GNSS-corrected GNSS solution, and the updated state vector estimate becomes $\hat{x}_k^+$. When updating the state vector estimate, an adaptive Kalman filter (AKF) is applied by adjusting the a priori knowledge about the overbounded measurement noise

covariance, as provided by the LAD-GNSS architecture. The derivation for the overbounded error model of the LAD-GNSS is defined in earlier work [2]. $VPL_{H0, MS}$ is then computed in a simplified form by (1),

$$VPL_{H0, MS} = k_{FFMD} \cdot \sigma_{\text{vert}, H0, MS} \quad (1)$$

where $\sigma_{\text{vert}, H0, MS}$ is the EKF-derived standard deviation of the vertical position error and k_{FFMD} is computed by the revised integrity fault tree. It should be noted that both the process noise covariance and the measurement noise covariance should be properly overbounded to compute the resulting position error covariance values that are bounded. When designing the EKF in this study, the process noise covariance is chosen manually by trial and error such that the state vector reaches a steady state with a desirable transient response with the provided overbounded measurement noise covariance. The resulting $VPL_{H0, MS}$ is verified by comparing the computed VPL and the position errors, as will be discussed in

Section. 3. However, to be exact, an overbounding study of the process noise and its verification must be conducted under dynamic flight conditions when deriving the $VPL_{H0, MS}$. This study will be conducted in later research.

2.3 Single-satellite vehicle failure hypothesis: H2, SSV

Section 2.1 presents the proposed monitoring strategy for a SSV failure as applied to the LAD-GNSS/onboard sensor integrated system. The inclusion of the onboard sensor failure hypothesis to the LAD-GNSS integrity/continuity fault tree (without changing the total requirement) places more of a burden on other failure hypothesis monitors. The integrity allocation of the SSV failure hypothesis is reduced from 1.4×10^{-7} to 7×10^{-8} as a result. The proposed monitoring concept utilizes both the ground monitor of the LAD-GNSS and the onboard monitor to capture the benefits from the integration of the onboard sensors. In the ground monitor, as a first step, signals are monitored through a comprehensive set of monitoring algorithms to detect a varied array of possible failures in the GPS SI [1,6]. The ground monitor, which includes signal quality monitoring (SQM), data quality monitoring (DQM), measurement quality monitoring (MQM), a multiple-reference consistency check (MRCC), Sigma-mean ($\sigma\mu$) monitoring, and a message field range test (MFRT), is designed to ensure a 7×10^{-6} /approach integrity level. The signals monitored from the ground monitor are then re-monitored in the onboard monitor with a lower a-priori probability to the point ensured by the ground monitor. As a result, the missed detection probability (P_{MD}) of the onboard monitor is reduced to 1×10^{-2} , which is calculated by dividing the allocated integrity risk (P_{HMI}) by the a-priori probability (P_{prior}), as shown in (2); thus, the corresponding VPL is also reduced.

$$\begin{aligned} P_{MD, \text{onboard monitor}} &= \frac{P_{HMI, SSV}}{P_{prior, GPS \text{ after ground monitoring}}} \\ &= \frac{7 \times 10^{-8}}{7 \times 10^{-6}} = 1 \times 10^{-2} \end{aligned} \quad (2)$$

In this study, the onboard monitor adopted the concept of the Weighted Least Square RAIM, as proposed for GPS-based precision approach by Walter and Enge [4]. The Weighted Least Squares RAIM (WLS RAIM) is a snapshot-based monitoring scheme which can detect a single measurement fault. It monitors the consistency of measurement residuals based on the position estimated by a navigation filter. The VPL under the SSV failure hypothesis consists of two components, as shown in (3). First, the position error bound is computed based on the bias induced by the satellite fault. Second, the random error term is computed using a sigma-multiplier (k_{MD}), which considers that monitors miss-detect such faults.

$$VPL_{SSV} = \mu_{SSV} + k_{MD} \cdot \sigma_{vert, SSV} \quad (3)$$

$$k_{MD} = Q^{-1}(P_{MD}) \quad (4)$$

In the case of EKF-based navigation, the position error bound (μ_{SSV}) is derived recursively from the EKF. Assuming that a failure has occurred in satellite i , μ_{SSV} at time t_k can be written as follows [5],

$$\mu_{SSV, k} = (I - K_k H_k) F_k \mu_{SSV, k-1} - K_k B_{SSV, k}, \quad (5)$$

where K_k is the Kalman gain, F_k is the state transition matrix, H_k is the measurement matrix, I is the identity matrix, and B_{SSV} is worst-case bias obtained from the WLS RAIM under the failure hypothesis for satellite i . As shown in (5), μ_{SSV} consists of the accumulated error due to the recursive process of the Kalman filter and the current error based on the worst-case bias ($B_{SSV, k}$). As the term $(I - K_k H_k) F_k$ is less than one, the accumulated error tends to converge over time. Finally, the VPL_{SSV} is calculated as shown in (3). In equation (3), σ_{SSV} is obtained from the a posteriori state error covariance, which is defined similarly to the fault-free protection level ($VPL_{H0, MS}$).

2.4 Onboard Sensor Failure Hypothesis: H3

To monitor the onboard sensor failure, the innovation-based fault monitoring method is applied [6]. The innovations are an intermediate product of the KF process, providing a simple test statistic under the INS failure hypothesis while assuming no fault with regard to other types of failures. The test statistic at time t_k can be expressed mathematically as

$$\alpha_k = z_k - H_k \hat{x}_k^-, \quad (6)$$

where z_k is the measurement from the monitored and corrected GNSS measurements from the LAD-GNSS corrections and $\hat{x}_k^-$ is the a priori state vector. Given that the error distributions are expected to follow Gaussian distributions under fault-free conditions, a test statistic should be normally distributed with a zero mean and variance with the covariance of the innovation ($C_{\alpha, k}^-$).

Here, $C_{\alpha, k}^-$ consists of the sum of the measurement noise covariance and the error covariance of the state estimates transformed into the measurement space, as shown in (7),

$$C_{\alpha, k}^- = H_k P_k^- H_k^T + R_k, \quad (7)$$

where P_k^- is the covariance of the state estimates and R_k is the measurement noise covariance. The threshold $T_{th,\alpha}$ is chosen such that the probability of false alarm (P_{FA}) is commensurate with the continuity requirement. The position error bound (μ_{H3}) for the INS failure case is calculated similarly to the method in (5), but with a difference in that the bias induced by the INS fault is reflected in a priori state estimate. Then, the value of μ_{H3} at time t_k can be formulated as follows:

$$\mu_{H3,k} = (I - K_k H_k) F_k \mu_{H3,k-1} + B_{H3,k} - K_k B_{H3,k} \quad (8)$$

Finally, the VPL for the H3 hypothesis is given by

$$VPL_{H3} = \mu_{H3} + k_{MD} \sigma_{vert,H3}, \quad (9)$$

Where σ_{H3} is the variance of the position error obtained from the a-posteriori state error covariance from the EKF.

3. VERIFICATION STUDIES WITH UAV FLIGHT TESTS

3.1 Flight Test Configuration

Figure 3 shows the KAIST LAD-GNSS Test-bed, which is aimed at meeting GBAS Category I (CAT-I) integrity performance. The multipath antennas setup (connected to the NovAtel ProPak-V3) mounted on the rooftop of this test-bed is similar to that of the GBAS Ground Accuracy Designator B (GAD-B) [3]. Thus, the GAD-B model is used for the multipath environment at this site. Using this test-bed, UAV flight tests were conducted at locations near the campus. These two sites are located about 4.1 km apart. The differential corrections and integrity information generated by the LAD-GNSS Test-bed are available to users anywhere on the campus.

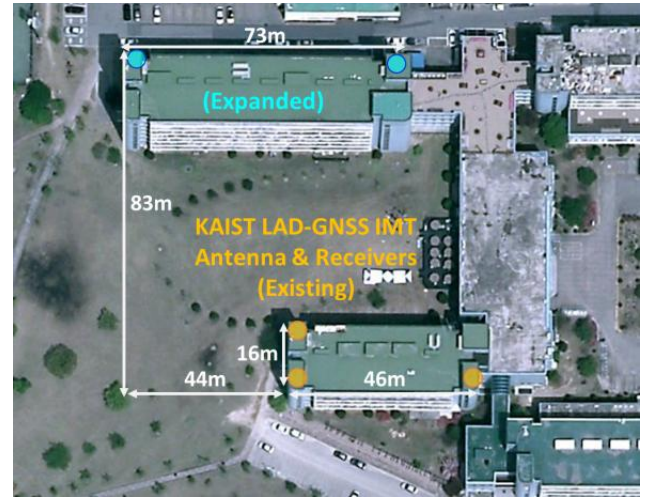


Figure 3. Configuration of the LAD-GNSS test-bed at KAIST



Figure 4. UAV platform for the flight tests

To perform the flight tests, an octocopter 110 cm in diameter, 35 cm in height, and 5.26 kg in weight was used as shown in Figure. 4. The vehicle has flight equipment to receive LAD-GNSS differential corrections and can estimate precise positions using the EKF and fly autonomously. This equipment includes a modem, a NovAtel FlexPak6 receiver, a NovAtel compact antenna, and Pixhawk controller with an internal INS sensor of the Pixhawk controller [7]. This octocopter performs flight using the navigation solution of the EKF and records GNSS and INS data in its data logger at the same time.

3.2 Results

In order to evaluate the performance of LAD-GNSS/onboard sensor integrated system, UAV flight tests were conducted. Figure 5 shows one of the trajectories from the flights. The blue line is the truth trajectory generated by employing the double difference of the carrier phase and wide-lane geometry-free observables [8]. As this method results in position error at or below the 10-cm level, the positions based on carrier-phase observables were assumed as true. In Figure 5, the red line represents the EKF-derived trajectory generated from

our integrated system. The standard deviation of the estimated position error was 0.15 m in the vertical direction. This value meets the intended accuracy requirement for the GBAS CAT-I.

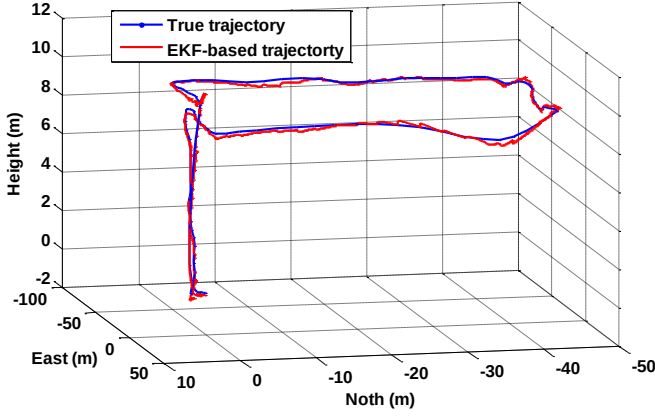


Figure 5. Flight test trajectories (True vs. EKF-estimated)

Using the collected data during flight tests lasting for approximately 4 minutes, the VPLs for each failure hypothesis were derived. First, $VPL_{H0, MS}$ is derived as shown in Figure 6.

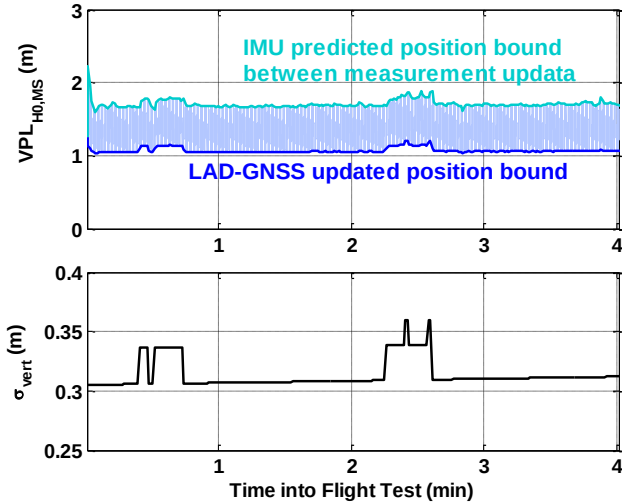


Figure 6. $VPL_{H0, MS}$ (top) and the standard deviation of the LAD-GNSS measurement error (bottom)

The upper plot in Figure 6 shows $VPL_{H0, MS}$ as computed with $K_{FFMD} = Q^{-1}\left(\frac{P_{FFMD}}{2}\right) = 5.73$, where one half of P_{FFMD} is allocated to the VPL computation. The upper boundary shows VPL during the propagation process between each measurement update, and the lower boundary shows VPL after the measurement updates. The bottom plot in Figure 6 shows σ_{vert} , applied as the

measurement noise covariance provided by the LAD-GNSS.

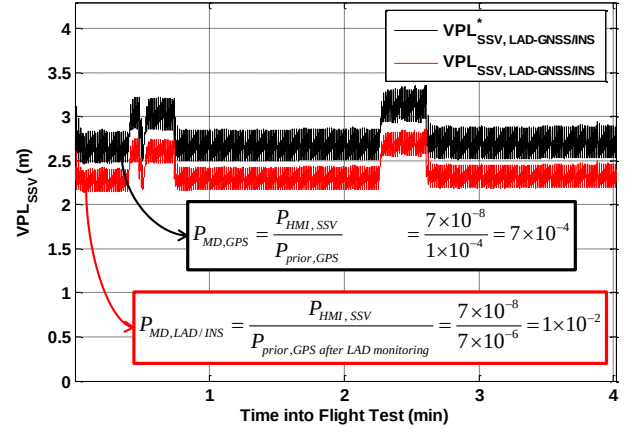


Figure 7. VPL_{SSV} computed from an onboard RAIM monitor (black curve) and the proposed ground-based/onboard combined monitor (red curve)

Figure 7 shows VPL_{SSV} derived from the LAD-GNSS/onboard sensor integrated system. Two different VPL_{SSV} curves represent the performance difference between the two monitoring approaches; one is the monitoring method which only utilizes the onboard RAIM monitor, and the other is the monitoring strategy proposed in this study that utilizes both the LAD-GNSS monitor and the onboard monitoring system. The black curve shows the VPL_{SSV} value when only the onboard RAIM monitor is applied. In this case, $P_{prior,GPS}$ is 1×10^{-4} , resulting in a corresponding P_{MD} of 7×10^{-4} . The red curve is computed from the proposed monitoring system. The monitored signals from the LAD-GNSS ground monitor provide the benefit of reducing the a-prior probability of a SV failure, resulting in a released P_{MD} of 1×10^{-2} . As a result, VPL_{SSV} from the proposed monitoring concept has smaller values, which would improve the system availability while reducing the burden of the LAD-GNSS monitors.

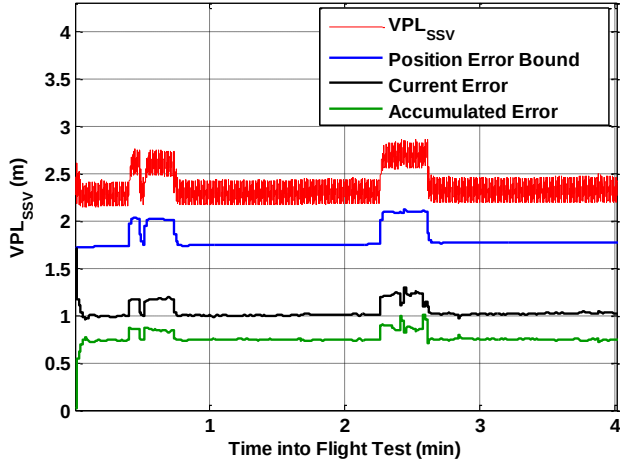


Figure 8. Term-by-term analysis of the VPL_{SSV}

Figure 8 represents the decomposed error terms of the VPL_{SSV} , showing the effects of recursive EKF positioning on the resulting VPL_{SSV} values. As explained in Section 2.3, the current position error is computed by the maximum bias derived from the onboard RAIM monitor. The black line shows the current error terms as a time series, while the accumulated errors due to the recursive positioning of the EKF are denoted by the green line. It is shown that the accumulated error grows during the initial operation periods but converges within several seconds after the beginning of the flight. The position error, which is the sum of these two components, is denoted by the blue line. The resulting VPL_{SSV} is then computed by applying the uncertainty of the position error under SSV faulty conditions with the corresponding P_{MD} .

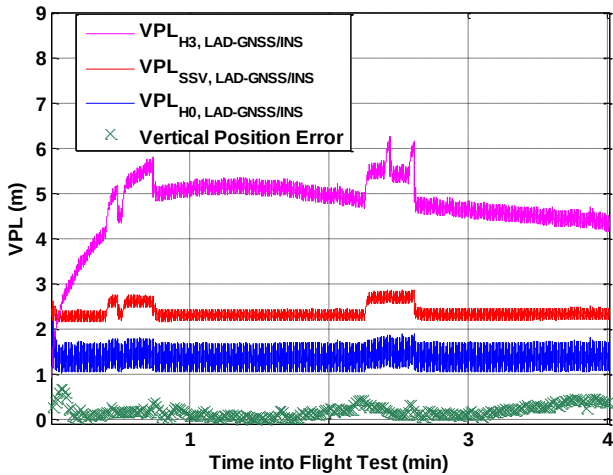


Figure 9. VPLs under different failure hypotheses

Figure 9 shows all of the VPLs under the fault free and different failure hypotheses. It is shown that the VPL_{H3} has the largest value for most of the operation time. It should be noted that the level of this VPL_{H3} can be reduced by using a high-grade INS. The resulting VPL is

chosen by taking the maximum VPL among the computed VPLs under various failure hypotheses as shown in (10). The VPL_{H1} in (10) is the VPL that assumes single-reference-receiver failures (H1). Since this study focuses on hypotheses that are mainly revised for the proposed integrated system, the derivation of the VPL_{H1} was not conducted in this study. However, it is expected that the VPL_{H1} would not exceed the VPL_{SSV} due to its distinctive monitoring method. Note that the VPL_{H1} is computed based on actual B-values, which is broadcasted to users in real-time, instead of worst-case undetected B-values [9].

$$VPL = \max(VPL_{H0,MS}, VPL_{H1}, VPL_{SSV}, VPL_{H3}) \quad (10)$$

It is shown that the vertical position error is properly bounded by the resulting VPL.

4. CONCLUSION

This paper presented the concept of a high-integrity navigation architecture for the integrated LAD-GNSS and an onboard navigation sensor system with an EKF-based navigation filter. The paper introduced preliminary but essential issues pertaining to a LAD-GNSS/onboard sensor integrated system, consisting of an integrity/continuity requirement design, fault monitoring strategies, and corresponding PL computation formulations.

Although one half of the total integrity requirement is allocated to the onboard sensor failure hypothesis, the proposed integrity fault trees guarantee the integrity of the system by adjusting the sub-allocations of fault modes and redesigning the monitoring methods. To meet the tight integrity allocation of a SSV failure, both the ground monitor of the LAD-GNSS and the onboard RAIM are applied to monitor satellite faults. In addition, an innovation-based fault-monitoring method is utilized to monitor onboard sensor failures.

In this paper, flight experiments were conducted with a multi-rotor platform, and INS is selected as an onboard sensor. Using data collected from flight tests, the VPLs under the fault-free and fault-mode hypotheses were derived to verify the performance of the proposed system. The results show that the VPL for the INS failure hypothesis has the largest value for the most of operation time. The level of this VPL can be reduced by using a high-grade INS and with improved monitoring methods for INS sensor failures. To improve the performance of the proposed framework, optimal allocation of the integrity and continuity budgets among the failure modes, overbounding INS sensor errors under dynamic conditions, and the design of advanced monitors for different types of onboard sensors will be considered in the future work.

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